

TECHNICAL ARCHITECTURE — PHASE 2

Technical Architecture — Phase 2

Events + Social Seed

v2.0 · Richard Uzor — May 2026 · Devstrike Digital Limited

01 — PHASE 2 ARCHITECTURAL ADDITIONS

Phase 2 adds four new architectural components: a background job worker, a cache layer, Google Calendar OAuth, and the Pystack payments integration. The social feed is introduced as a standard REST API with no real-time requirement yet.

02 — NEW COMPONENTS

Component	Technology	Purpose
Job Worker	Separate Railway container (BullMQ + Redis)	Async tasks: email sends, quote card generation, webhook processing, report threshold checks
Cache Layer	Upstash Redis	Rate limiting, hot feed data (5-min TTL), reaction counts, session cache

Component	Technology	Purpose
Calendar OAuth	Google Calendar API + NextAuth.js Google provider	Account users save upcoming events to Google Calendar
Payment SDK	Paystack Node.js SDK	Event ad subscriptions, donations, verified profile fees
PWA	next-pwa service worker	App installability, offline archive read, push notification registration
Push notifications	Firebase Cloud Messaging (FCM)	Event reminders, 'pastor posted', 'question answered' notifications

03 — NEW API ENDPOINTS (PHASE 2)

Method	Endpoint	Auth	Description
POST	/api/v1/payments/initiate	Account	Initiate Paystack payment — ad, donation, verification
POST	/api/v1/payments/webhook	Public	Paystack webhook receiver — validates HMAC-SHA512
POST	/api/v1/donations/initiate	Public	Initiate voluntary donation
POST	/api/v1/events/:id/calendar	Account	Trigger Google Calendar event creation
POST	/api/v1/feed/posts	Account	Create public feed post
GET	/api/v1/feed	Account	Get personalised feed — following + trending
POST	/api/v1/feed/:id/react	Public	React to feed post
POST	/api/v1/feed/:id/comment	Account	Comment on feed post
POST	/api/v1/pastors/:id/follow	Account	Follow a pastor
POST		Account	Register FCM device token

Method	Endpoint	Auth	Description
	/api/v1/notifications/register		
GET	/api/v1/users/me/feed	Account	Get following-only feed

04 — PAYSTACK INTEGRATION ARCHITECTURE

All payments follow a strict three-step flow to ensure webhook reliability and prevent double-processing.

1. Initiate

- Client calls POST /api/v1/payments/initiate with type, amount, metadata.
- API creates a pending payment record in DB with Paystack reference.
- API returns Paystack checkout URL. Client redirects.

2. Process

- User completes payment on Paystack. Paystack fires webhook to /api/v1/payments/webhook.
- API validates HMAC-SHA512 signature. Rejects if invalid.
- API retrieves pending payment by reference. Updates status to 'success' or 'failed'.

3. Activate

- On success: activates relevant feature (Sponsored badge, verified check, etc.).
- Sends receipt email via Resend.
- Idempotent: duplicate webhook fires are safely ignored (status already updated).